
Vincenzo Ruotolo

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Senior Software Engineer

Software Engineer with experience in consultancies, start-ups and as a contractor. Deep expertise in implementing robotics architecture, core algorithm modules, autonomous systems, scalable cloud app and ETL pipelines for telemetry and test data. Additional strengths in embedded systems, ML/AI, DevOps and automated testing. Proven track record delivering production-grade software for automotive ADAS, cloud-native APIs, and robotic platforms. Experienced in leading and coordinating cross-functional agile teams.

WORK EXPERIENCE

Grid Dynamics

Software developer | Google • Full-time

Guadalajara, Mexico • 05/2026 - Present

C++ Software Engineer at Grid Dynamics, contributing to Google's TensorFlow codebase, responsible for diagnosing and resolving defects across TensorFlow's C++ core to support the stability and performance.

- Investigate and resolve critical C++ defects (out-of-bounds read/write, memory leaks and other memory-safety issues) in TensorFlow codebase
- Perform root-cause analysis using debugging tools, sanitizers, and profilers to isolate hard-to-reproduce, low-level bugs affecting stability and correctness

Amaris

Senior software developer | Caterpillar • Full-time Mexico • 01/2026 - 04/2026

Develop software components for sensors like GPS, in a microservice architecture in C++.

- Design and develop core GPS module, within a microservice platform
- Design and implement testing strategies

Autonomous Robotic Systems

Senior software developer • Contractor

Mexico • 06/2024 - 12/2025

Design, develop and deploy a flexible robotics system for different platforms (UGV / UAV / UUV), aiming to software modularity and scalability.

- Develop and roll out core algorithm modules (Sensors, GNC, SLAM) using efficient, real-time C++ and utilities modules (remote teleoperation, telemetry, diagnostics, infrastructure) using Python, within ROS2 middleware
- Develop embedded firmware, GPU acceleration, hardware drivers and system integration
- Lead and coordinate a cross-functional agile team of 8 engineers

Luxoft (DXC)

Software automation developer | Ford • Full-time

Mexico • 05/2023 - 10/2025

Engineered an autonomous testing framework in Python for Ford's advanced driver aid (ADAS L2+), enhancing check comprehensiveness and shortening evaluation duration.

- Built a Python automation (ADAS L2+), combining orchestration with ETL to ingest and replay vehicle telemetry for deterministic validation

SKILLS

OS: Ubuntu, Kali, Windows

Programming: C++, C, Python

Data: Pandas, SQL, NoSQL, FastAPI, Kafka, Airflow, Spark

Robotics: ROS2, micro-ROS, Gazebo, Isaac Sim

Embedded: FreeRTOS, IoT, RPi

ML/AI: Scikit-learn, Pytorch, OpenCV, YOLOv8

Parallel Computing: CUDA

DevOps: Git, GitHub Actions, Ansible, Docker, Agile/Scrum, CI/CD, Terraform, AWS, Kubernetes

Testing: TDD, pytest/unittest, gtest/gmock

Build tools: Make, CMake, Bazel

Languages: Italian, English, Spanish

EDUCATION

Master's Degree in Mechatronic Engineer

Politecnico di Torino
10/2018 - 04/2021

Thesis: Optimal Trajectory Generation via MPC for a Four-Wheel Steering UGV

Bachelor's Degree in Aerospace Engineer

Politecnico di Torino
10/2014 - 07/2018

Thesis: Thales Alenia Space internship

- Implemented CAN / UDS / TCP / UDP adapters for trace capture, injection and end-to-end replay
- Developed unit, integration, smoke, sanity suites and Python harnesses that shifted many checks off HIL into CI, reducing regression time

Cosmico

Software developer • Contractor

Remote, Italy • 01/2023 - 12/2023

Built a cloud-native backend for automated technical interview evaluation using FastAPI and SQLAlchemy, implementing candidate management, pre-formed assessments and result aggregation.

- Built cloud-native backend (FastAPI + SQLAlchemy) for automated interview evaluation; implemented candidate management and assessment orchestration
- Implemented ETL pipelines to ingest candidate submissions (code, logs, recordings), normalize and persist artifacts (S3), and produce queryable datasets for analytics and recruiter dashboards
- Integrated ML/AI inference for automated scoring (code quality, tests, plagiarism detection) and designed retraining pipelines using labeled outcomes and feedback loops

Luxoft (DXC)

Software developer | BMW • Full-time

Turin, Italy • 01/2022 - 04/2023

C++ software developer for Adaptive AUTOSAR application layer in BMW Autonomous Driving (ADAS L2+)

- Implemented ETL pipelines to ingest candidate submissions (code, logs, recordings), normalize and persist artifacts, and produce queryable datasets for analytics and recruiter dashboards
- Implemented modern C++ safety critical modules for Adaptive AUTOSAR (ARA::COM); designed efficient serialization/deserialization for ECU data flows
- Created C++ unit and integration tests and Python bindings for automated validation

PROJECTS

ESA - Space Resources Challenge

10/2021 - 04/2023

DRAFT

The challenge aims to develop an autonomous system of UGV and UAV, collaborating in a unknown space environment. Led the robotics architecture, core module development, integration, and DevOps.

LDC - Leonardo Drone Contest

10/2019 - 10/2021

DRAFT PoliTO

The contest aims to develop autonomous drones in contest of GPS-denied condition. Designed the high-level robotics architecture, led component integration and simulation.